

OWR Conservation Technology Strategy — Executive Summary

Okapi Wildlife Reserve (Ituri, DRC) | 2026–2030

Wildlife Conservation Society (WCS) — Conservation Technology Unit

The OWR Conservation Technology Strategy provides a five-year roadmap (2026–2030) to modernize field operations, strengthen law enforcement monitoring, and scale ecological and social monitoring through integrated sensors, data systems, and local capacity building. The strategy prioritizes rapid data lifecycles, evidence-based governance, and sustainable, community-engaged technology deployment to protect okapi and other key species.

Key Objectives

- Digitize patrol and incident reporting to eliminate paper transcription and enable near-real-time CCOPS visibility.
- Integrate multi-source sensors (camera traps, telemetry, acoustic, eDNA, satellite alerts) into EarthRanger and SMART.
- Automate camera-trap and canopy-loss processing with AI and remote-sensing pipelines to reduce analysis lag.
- Build local training and a train-the-trainer program to ensure long-term technical sustainability.

Core Actions & Timeline

- Phase 1 (2026): Issue SMART Mobile devices, retire paper forms; launch CCOPS dashboards.
- Phase 2 (2027): Deploy Gundi middleware and full sensor ingestion into EarthRanger.
- Phase 3 (2027–2028): Integrate Wildlife Insights AI and automate GFW alerts; expand drone/eDNA pilots.
- Phase 4 (2028): Publish OWR Impact Dashboard and public story maps; scale community monitoring.

Priority KPIs

- 100% digital patrol logging; zero paper transcriptions by end Phase 1.
- 80% reduction in camera trap processing bottleneck post-AI integration.
- 80% of patrol reports ingested digitally within 24 hours (data latency).

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Full strategy and references: see OWR _ConsTec _Strategy.docx in the repository.